## Supporting Information

# Structural and Transport Properties of Norbornene Functionalized Polyvinyl Alcohol 'Click' Hydrogel: A Molecular Dynamics Study

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Table S1. System Details

| Polymer name | Initial box dimension ($nm^3$) | Final box dimension ($nm^3$) | Total number of polymer molecules | Total number of water molecules | Total number of molecules |
|---|---|---|---|---|---|
| PVA | 18.427×18.427×18.427 | 6.010×6.010×6.010 | 1124 | 27432 | 28556 |
| PVA-N | 18.427×18.427×18.427 | 7.054×7.054×7.054 | 1444 | 44636 | 46080 |
| PVA-NT | 18.656×18.656×18.656 | 6.768×6.768×6.768 | 1732 | 38700 | 40432 |
| PVA-NTC | 26.043×26.043×26.043 | 8.747×8.747×8.747 | 6352 | 80484 | 86836 |

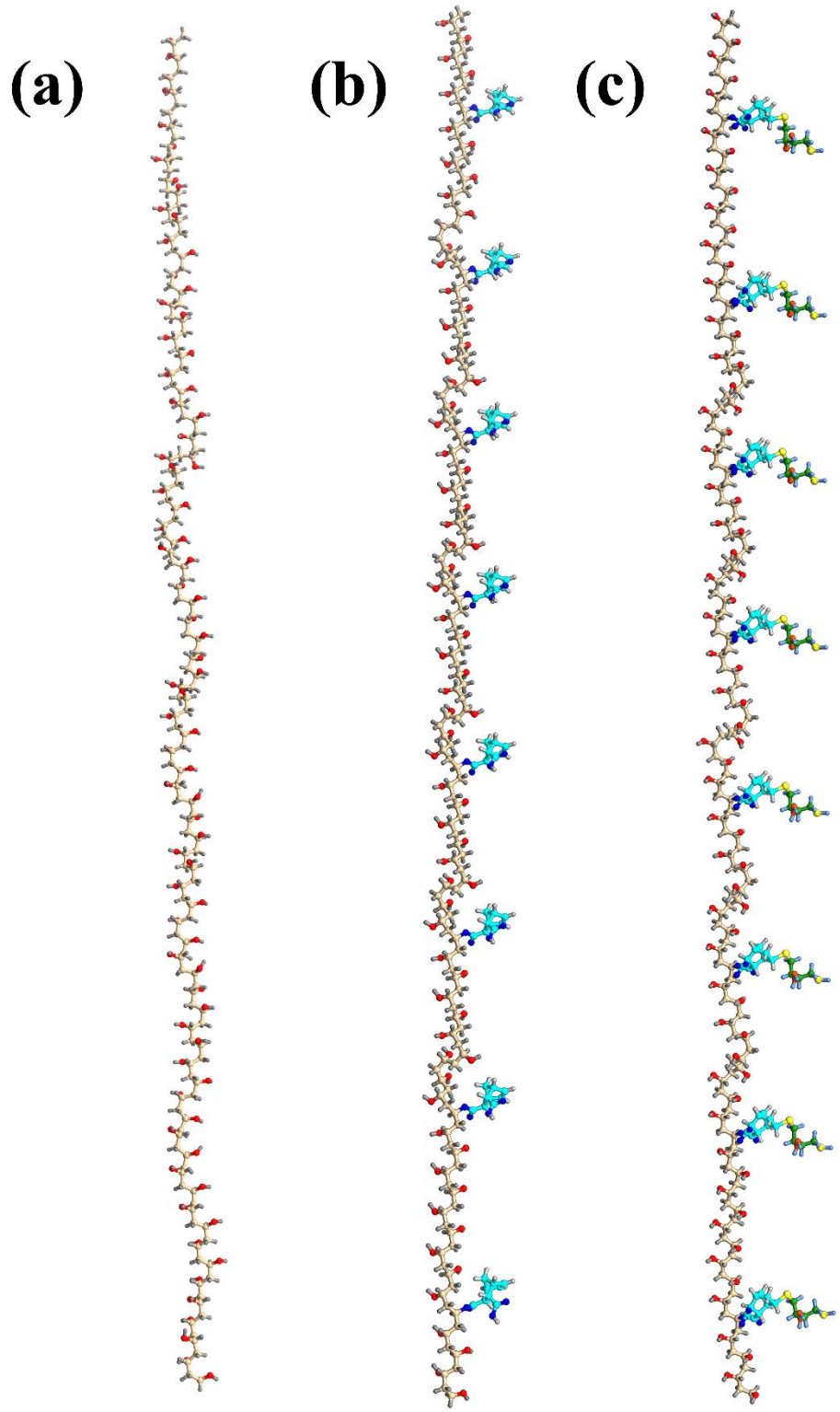



**Figure S1.** The initial geometry (a) PVA, (b) PVA-N, and (c) PVA-NT polymer chain

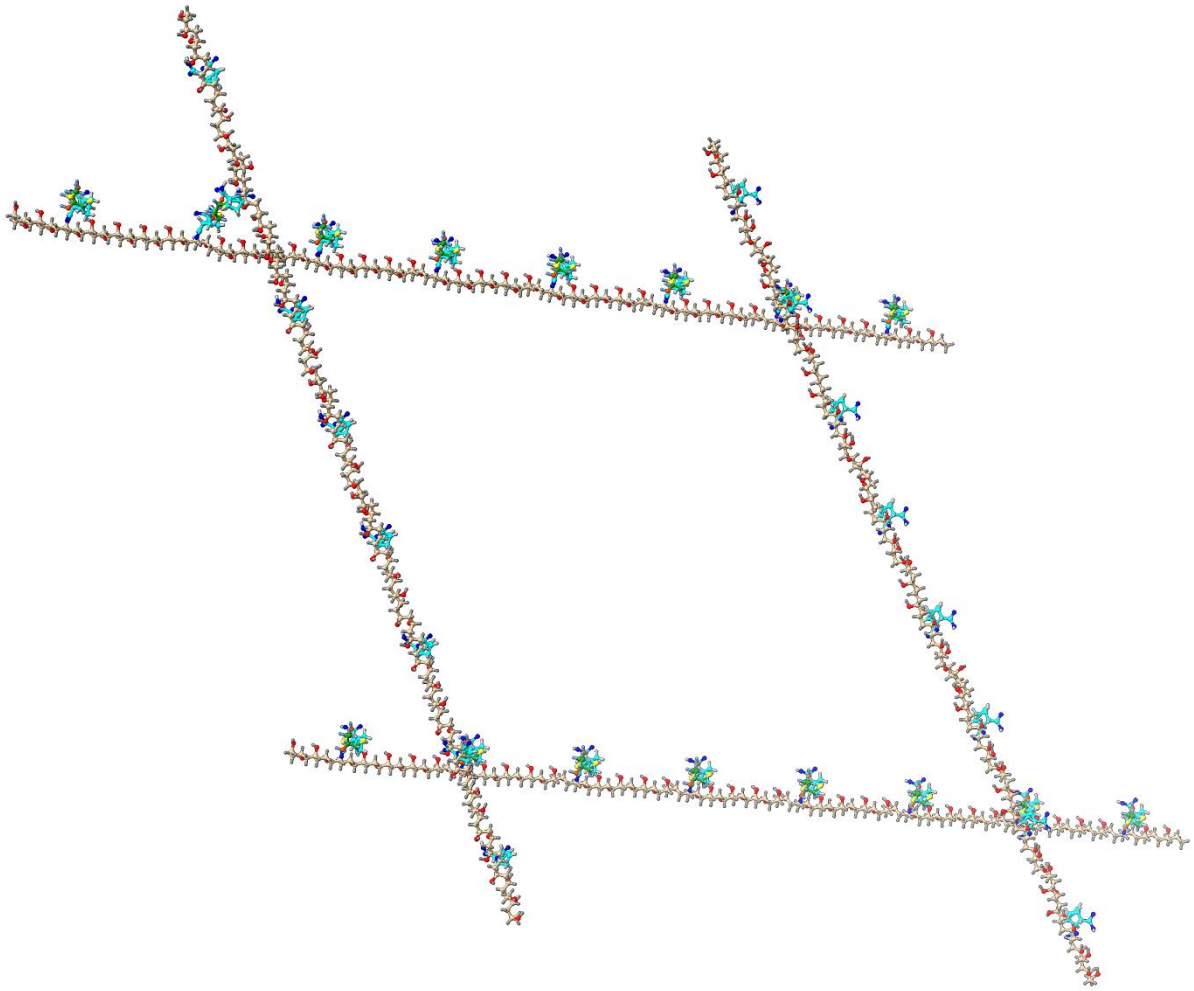


**Figure S2.** The initial geometry of PVA-NTC polymer chains

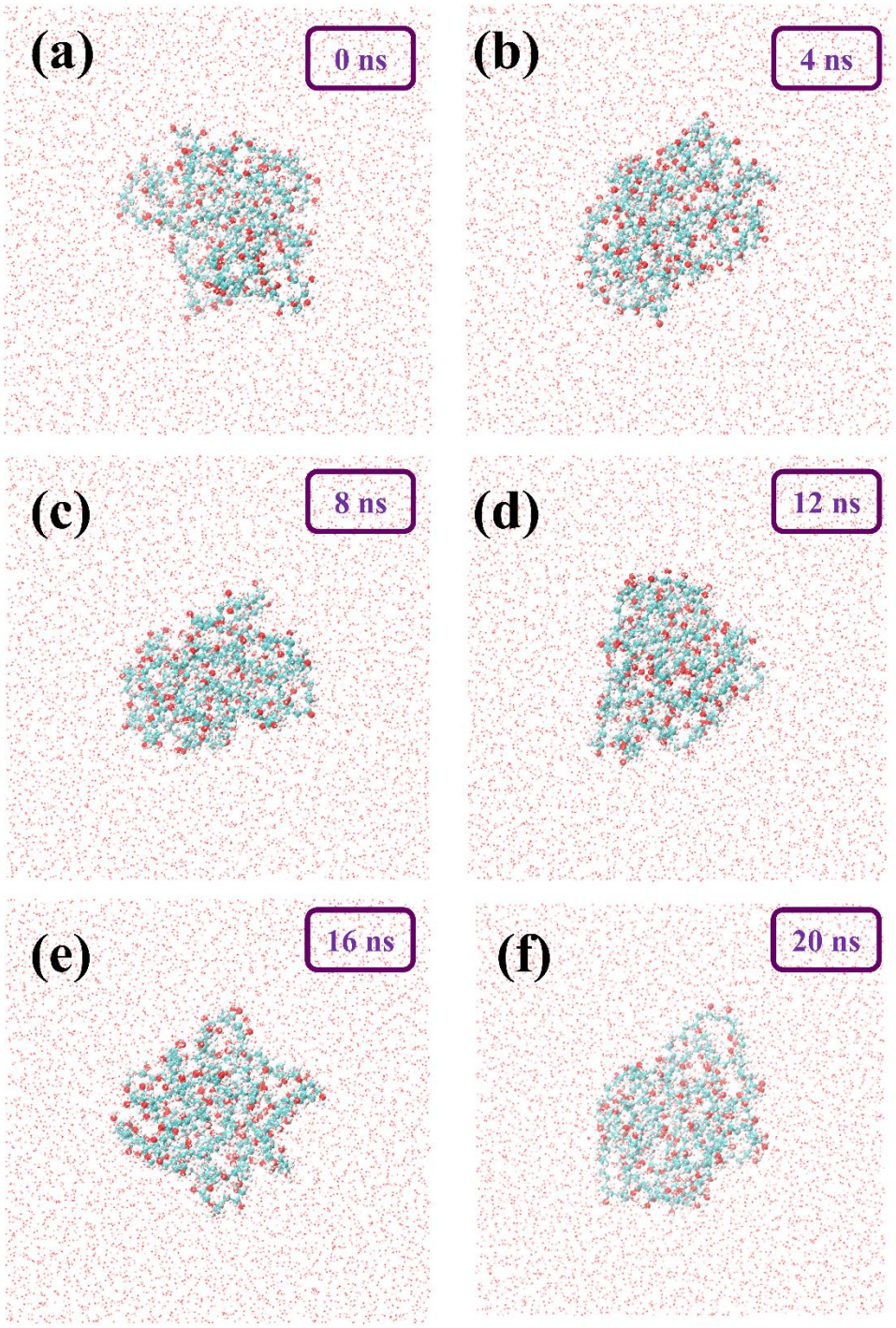



**Figure S3.** Simulation snapshot of PVA hydrogel at the different time intervals (a) 0 ns (b) 4 ns (c) 8 ns (d) 12 ns (e) 16 ns (f) 20 ns

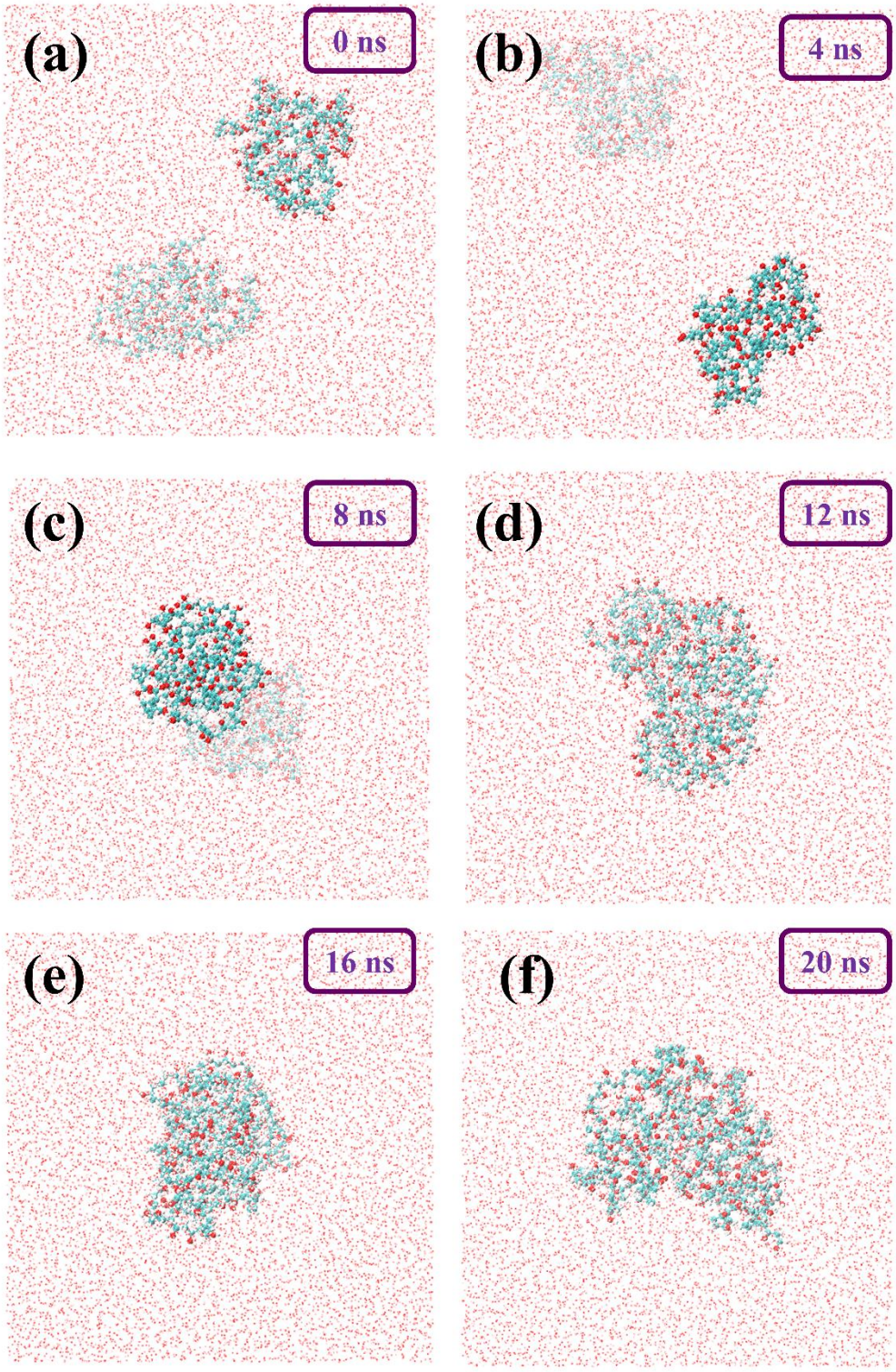



**Figure S4.** Simulation snapshot of PVA-N hydrogel at the different time intervals (a) 0 ns (b) 4 ns (c) 8 ns (d) 12 ns (e) 16 ns (f) 20 ns

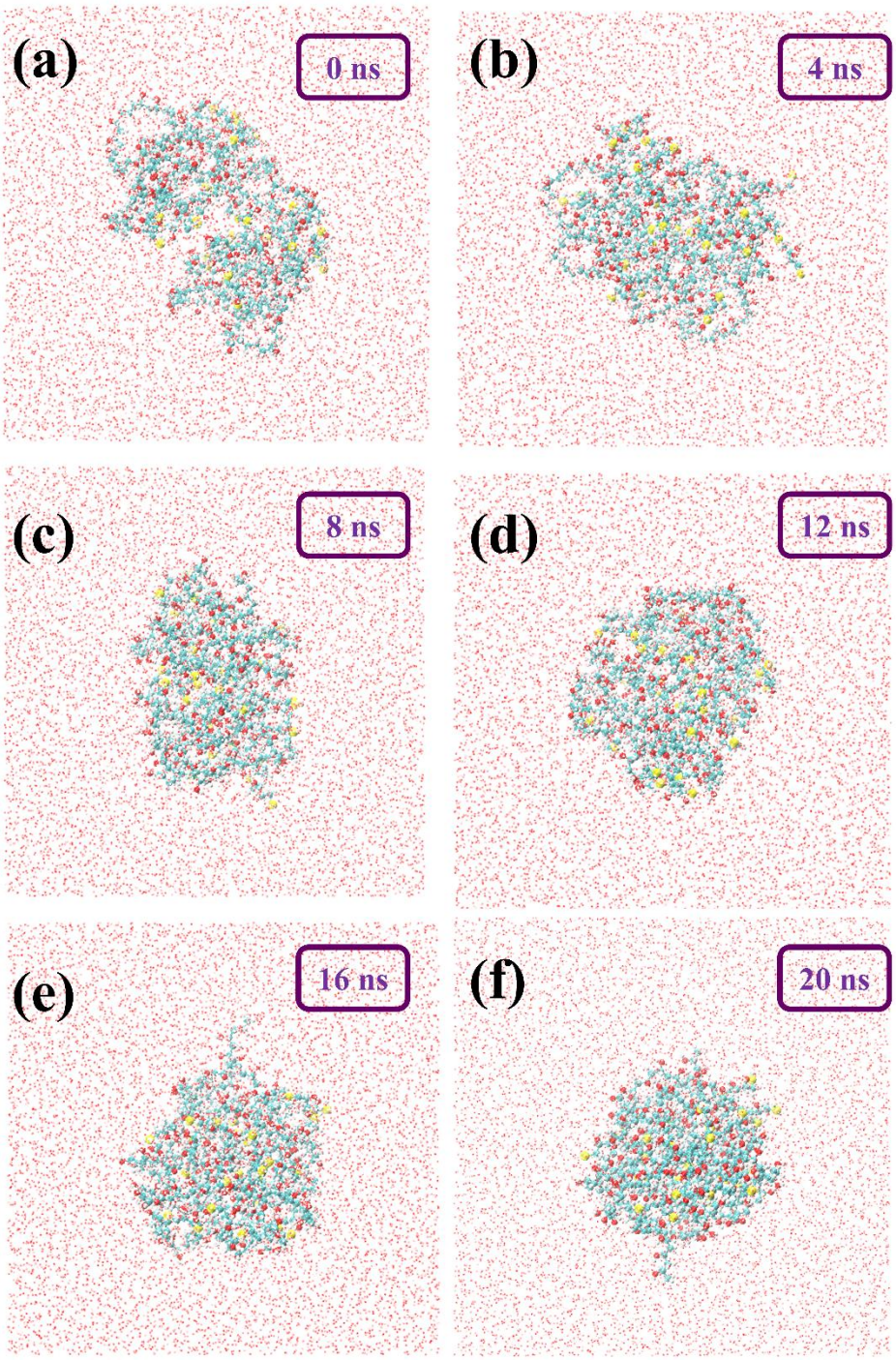



**Figure S5.** Simulation snapshot of PVA-NT hydrogel at the different time intervals (a) 0 ns (b) 4 ns (c) 8 ns (d) 12 ns (e) 16 ns (f) 20 ns

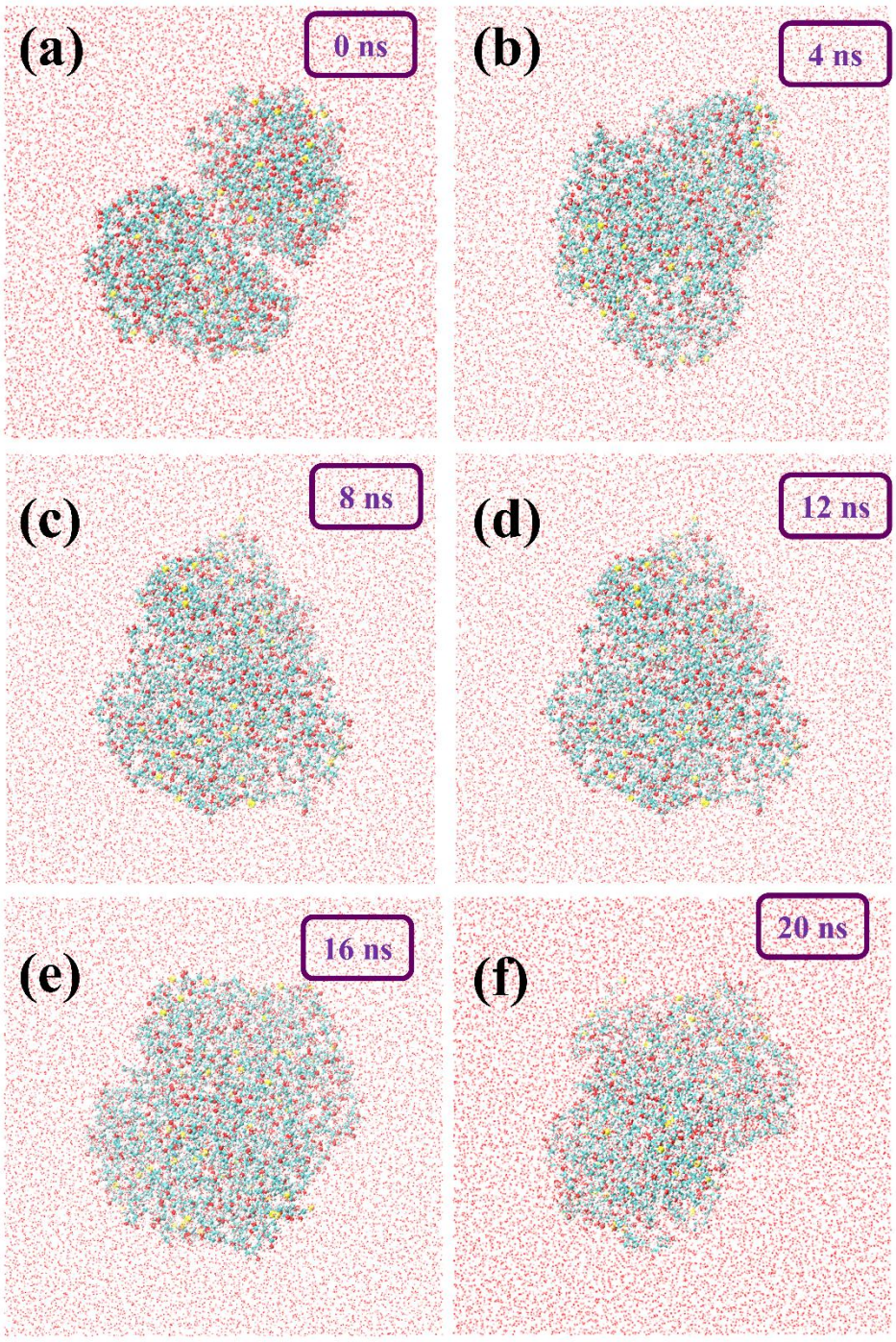



**Figure S6.** Simulation snapshot of PVA-NTC hydrogel at the different time intervals (a) 0 ns (b) 4 ns (c) 8 ns (d) 12 ns (e) 16 ns (f) 20 ns